

Ronit Saha

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SKILLS SUMMARY

Languages: Python, TypeScript, Java, Rust, Kotlin, JavaScript, C++, SQL
Frameworks & Libraries: FastAPI, SQLAlchemy (async), Celery, Next.js 15, React 19, Node.js, Express, Spring Boot, Tailwind CSS, ONNX Runtime Web, deck.gl, tree-sitter
Databases: PostgreSQL, PostGIS, MongoDB, MySQL, Redis, SQLite / Room, Firestore, Alembic migrations
Cloud / DevOps: Docker, Docker Compose, GitHub Actions (CI/CD), Linux, Vercel, Render
Developer Tools: Git, pytest, Vitest, Playwright, Criterion, Ruff, mypy, Postman, Cargo
AI / ML: On-device model inference, ONNX Runtime Web, WebGPU / WASM, SegFormer segmentation, computer vision, LLM applications (LLaMA 3.1 via Groq), prompt engineering, Faster-Whisper
Core Subjects: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, System Design

PROJECTS

TerraMind AI – Earth Intelligence Platform | [GitHub](#) Jul 2026 – Present

- Built a satellite-analysis backend as sole author across 90 commits and 562 files, layering versioned routers over services and seven repositories behind a single async Unit of Work, so every request commits or rolls back as one transaction.
- Moved minutes-long segmentation off the request path onto a Celery job platform tracked as database rows through a job lifecycle, with a lineage record written back on completion — no analysis can succeed without being recorded.
- Built the client globe as its own rendering engine with an adaptive governor stepping visual quality through a hysteresis band on a rolling frame-time average — degrading quality instead of dropping frames — behind provider interfaces backed by ~50 pytest modules and CI gated on Ruff and mypy.

Tech Stack: Python 3.11, FastAPI, SQLAlchemy (async), PostgreSQL + PostGIS, Celery, Redis, Alembic, React 19, deck.gl, Docker Compose

PratiBimb – On-Device Visual Perception for Browser Agents | [GitHub](#) Sep 2026 – Present

- Lead engineering on a Smart India Hackathon 2026 entry (PS 26171, ISRO / Department of Space): a browser agent running vision models on-device, replacing sensitive spans with typed placeholders that preserve field layout, role and data type while removing every identifying character.
- Diagnosed and fixed a PNG-to-tensor preprocessing mismatch that shifted 16–34% of detections, achieving byte-identical parity with the Python reference without retraining.
- Shipped a Manifest V3 extension validated on Chrome 153 and Edge 153, maintaining a TypeScript monorepo of 274+ commits and 795 passing tests, and reviewing collaborator pull requests as code owner.

Tech Stack: TypeScript, ONNX Runtime Web, WebGPU / WASM, Chrome Manifest V3 (WXT), Vitest, Playwright, npm Workspaces

Cartograph – Cross-Language Code Architecture Engine (Open Source) | [GitHub](#) Sep 2026 – Present

- Contributed 9 merged pull requests in Rust to a static-analysis engine that links TypeScript, TSX and Python into one cross-language call graph, with evidence and confidence on every edge.
- Implemented a tracing-layer redaction system centralizing duplicated secret-detection logic (106 lines removed, 12 added), stripping token shapes, private keys and bearer headers from every log line.
- Eliminated a flaky concurrency defect by isolating shared test fixtures, cutting failures from 5 in 12 concurrent runs to 0 in 16, and shipped an evidence-bundle module covering 32 unit tests.

Tech Stack: Rust, tree-sitter, petgraph, redb, tokio, rayon, LSP, Model Context Protocol, Criterion, Cargo Workspaces

Stealth F.R.I.D.A.Y – Local-First Voice Agent — Dashboard & Automation Layer | [GitHub](#) May 2026

- Built the observability and automation layer around a shared local voice-agent core: a Next.js 15 dashboard, its typed WebSocket hook, three control API routes, the automation-engine tools and the macOS-to-Windows port.
- Made a long-running local process observable by publishing complete state snapshots rather than deltas over a WebSocket bridge, so a dashboard opened mid-command converges on the next message with no replay log.
- Owned reconnection with backoff inside a single typed socket hook rendering live agent state, last command, active module and system health.

Tech Stack: Python 3.11, Next.js 15, TypeScript, WebSocket, Faster-Whisper, Groq LLaMA 3.1

CERTIFICATIONS

- Developing Back-End Apps with Node.js and Express — IBM, offered through Coursera | Jul 2026
- DSA Placement Bootcamp: Master Data Structures & Algorithms for Industry — Grade O, Certificate of Merit | Centre for Professional Enhancement, LPU | Aug 2026
- Programming in Java — iamneo (an NIIT Venture), LPU | May 2026

EDUCATION

Lovely Professional University Bachelor of Technology in Computer Science and Engineering — School of AI and Emerging Technologies St Joans School Senior Secondary (Class XII) — 78%	Punjab, India Aug 2024 – Present West Bengal, India 2024
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